Limit-sketchable infinity categories

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Limit sketches

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Example. Let A be the small category generated by the square α:

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$$\begin{array}{ccc} 0 & \longrightarrow & 1 \\ \downarrow & \alpha & \downarrow \\ 2 & \longrightarrow & 3 \end{array}$$

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$$A \xrightarrow{\quad \mathbf{F} \quad} \mathbf{Set}$$

$$\begin{tikzcd} 0 \arrow[r, dashed] \arrow[d, dashed] \arrow[dr, phantom, "\alpha"] & 1 \arrow[d] \\ 2 \arrow[r] & 3 \end{tikzcd} \quad \longmapsto \quad \begin{tikzcd} A \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & C \arrow[d] \\ B \arrow[r] & D \end{tikzcd}$$

A model **F** of the sketch $(A, \{\alpha\})$ is a pullback of sets

Presentability

A category C is **accessible** if it is locally small and there exists a regular cardinal κ such that:

- ☑ C admits κ-*filtered* colimits.
- ☑ There is a small full subcategory of κ-*compact objects* that generates C under κ-*filtered* colimits.

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Presentable	**Accessible but not presentable**	**Not accessible**
Set	Fields	Top
Grp	Hilbert spaces	
sSet		
RMod		

Higher categories

Informally, an ∞-**category** (or $(\infty, 1)$-**category**) has objects and:

- ☑ n-morphisms between $(n-1)$-morphisms for all $n \geq 1$,
- ☑ Composition, identities and associativity of n-morphisms weakly up to a $(n+1)$-morphism for all $n \geq 1$.
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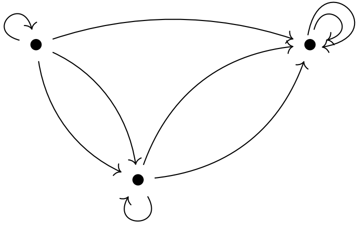

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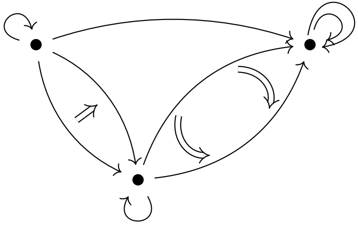

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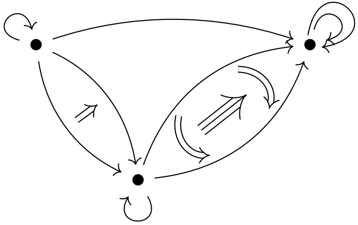

Representation theorem

We say that a category is **limit-sketchable** if it is equivalent to the category of models of some limit sketch.

Theorem

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Goal

Generalize this result to ∞-categories

Plan

1. Limit ∞-sketches
2. Presentable ∞-categories
3. Representation theorem

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Limits and colimits

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- The **constant diagram** $\underline{x} : \mathcal{I} \to \mathcal{C}$ over an object $x \in \mathcal{C}$ sends everything in $\mathcal{I}$ to x and its higher identities.

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- A **limit cone** of $D : \mathcal{I} \to \mathcal{C}$ is a cone $\ell : \underline{\lim D} \Rightarrow D$ such that, for any $x \in \mathcal{C}$, ℓ induces an equivalence

$$\mathrm{Hom}_{\mathcal{C}}(x, \lim D) \xrightarrow{\sim} \mathrm{Nat}_{\mathrm{Fun}(\mathcal{I},\mathcal{C})}(\underline{x}, D).$$

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Cocones and **colimit cocones** are defined as cones and limit cones in the opposite ∞-category.

Limit ∞-sketches

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Let $\mathcal{C}$ be a $\mathcal{L}$-complete ∞-category. A **model** of a limit ∞-sketch $\Sigma = (\mathcal{A}, \mathcal{L})$ in $\mathcal{C}$ is a functor $F : \mathcal{A} \to \mathcal{C}$ which sends each cone in $\mathcal{L}$ to a limit cone in $\mathcal{C}$.

$$\mathrm{Mod}(\Sigma, \mathcal{C}) \coloneqq \infty\text{-category of models of } \Sigma \text{ in } \mathcal{C}$$
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We say that an ∞-category is **limit ∞-sketchable** if it is equivalent to the ∞-category of models of some limit ∞-sketch.

An ∞-sketch $(\mathcal{A}, \mathcal{L})$ is **normal** if all the cones of $\mathcal{L}$ are limit cones.

Examples

- **Algebraic theories** (Rosicky 2007): Monoid objects (A_∞-spaces), commutative monoid objects (E_∞-spaces), group objects (∞-groups), R-modules, ...

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- Univalent category objects (Complete Segal spaces) and univalent operad objects (Complete Segal dendroidal spaces)
- ∞-sheaves $\simeq$ hypercomplete ∞-topos

Examples: Category objects

Let $\mathcal{C}$ be a complete ∞-category, and α_n be the cone over Δ^{op} with apex and diagram for all $n \in \mathbb{N}$:

$$\begin{tikzcd}
 & & & & {[n]} \arrow[dllll, dashed, bend right] \arrow[dll, dashed, bend right] \arrow[d, dashed] \arrow[drr, dashed, bend left] \arrow[drrrr, dashed, bend left] & & & & \\
{[1]} \arrow[dr] & & {[1]} \arrow[dl] \arrow[dr] & & {[1]} \arrow[dl] & & {[1]} \arrow[dr] & & {[1]} \arrow[dl] \\
 & {[0]} & & {[0]} & & \overset{(n)}{\cdots} & & {[0]} &
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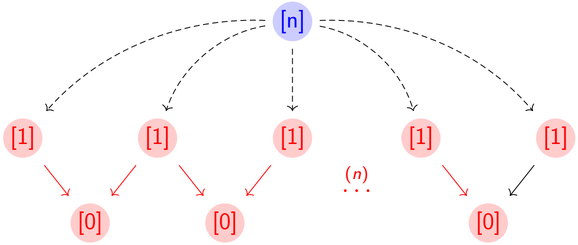

Then $\Sigma = (\Delta^{\mathrm{op}}, \{\alpha_n \mid n \in \mathbb{N}\})$ is a limit ∞-sketch, and a model $F : \Delta^{\mathrm{op}} \to \mathcal{C}$ is a simplicial object in $\mathcal{C}$ such that

$$F_n \xrightarrow{\sim} F_1 \times_{F_0} F_1 \times_{F_0} \overset{(n)}{\cdots} \times_{F_0} F_1. \qquad \textit{(Segal condition)}$$

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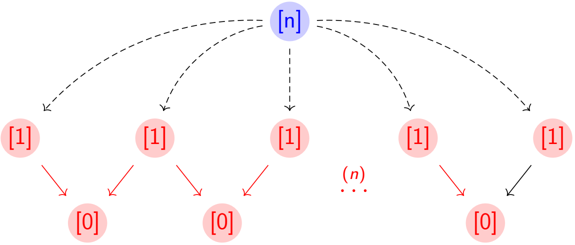

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$$\mathrm{Mod}(\Sigma, \mathcal{C}) \simeq \textbf{Category objects on } \mathcal{C}$$

$$\mathrm{Mod}(\Sigma) \simeq \textbf{Segal spaces}$$

Examples: Univalent category objects

Let $\mathcal{L}_S$ be the set of cones of the previous example, and β be the cone over Δ^{op} with apex and diagram:

$$
\begin{tikzcd}
& & {[0]} \arrow[dll, dashed] \arrow[drr, dashed] & & \\
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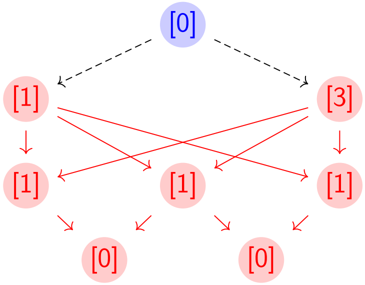

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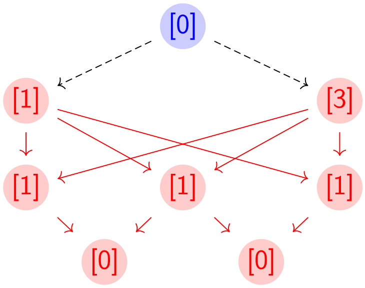

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$\mathrm{Mod}(\Sigma, \mathcal{C}) \simeq$ **Univalent category objects on** $\mathcal{C}$

$\mathrm{Mod}(\Sigma) \simeq$ **Complete Segal spaces**

Plan

1. Limit ∞-sketches

2. Presentable ∞-categories

3. Representation theorem

Filtered colimits and compact objects

Let κ denote a regular cardinal and $\mathcal{C}$ an ∞-category.

- A space $X \in \textbf{Spaces}$ is κ-**small** if $\pi_0(X)$ and $\pi_n(X, x)$ are κ-small sets for all $n > 0$ and $x \in X$.

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- An object $x \in \mathcal{C}$ is called κ-**compact** if the hom-space functor $\mathrm{Hom}_{\mathcal{C}}(x, -) : \mathcal{C} \to$ **Spaces** preserves κ-filtered colimits.

Accessibility

Definition (1)

An ∞-category $\mathcal{C}$ is **accessible** if it is locally small and there is a regular cardinal κ such that:

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Theorem (Joyal 2008, Lurie 2009). (1) $\iff$ (2)

Presentability

Definition

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Example

- The nerve of any presentable 1-category is presentable.
- The ∞-category of homotopy types **Spaces** is presentable.
- If $\mathcal{A}$ is a small ∞-category and $\mathcal{C}$ is a presentable ∞-category, then $\mathrm{Fun}(\mathcal{A}, \mathcal{C})$ is presentable.
- Any ∞-topos is presentable.

Plan

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Representation theorem

Theorem

(a) *For each limit ∞-sketch Σ, $\mathrm{Mod}(\Sigma)$ is presentable.*

(b) *For any presentable ∞-category $\mathcal{C}$, there is a normal limit ∞-sketch Σ such that $\mathcal{C} \simeq \mathrm{Mod}(\Sigma)$.*

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Corollary

(a) *An ∞-category is presentable if and only if it is limit ∞-sketchable.*

(b) *If $\mathcal{C}$ is presentable and Σ is a limit ∞-sketch, then $\mathrm{Mod}(\Sigma, \mathcal{C})$ is presentable.*

(c) *For each limit ∞-sketch Σ, there exists a normal limit ∞-sketch Θ such that $\mathrm{Mod}(\Sigma) \simeq \mathrm{Mod}(\Theta)$.*

Work in progress

- **Generalization:** A sketch is a limit sketch with a set of cocones which are sent to colimit cocones by any model. There is the following result due to Lair (revisited later by Makkai-Paré and Adámek-Rosický):

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Can this result be generalized to the ∞-categorical setting?

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- Formalize this work with a proof assistant which supports synthetic ∞-categories like `rzk`.

References

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Limit-sketchable infinity categories

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